

the collector side.

Similarly, a green-red-green glow means that the transistor is pnp type and the pin associated with the red LED is the base. To identify the emitter and collector, simply press switch S1 and

observe red LEDs associated with the already glowing green LEDs on the sides. The LED glowing with a high intensity indicates the emitter side, while the low-intensity LED indicates the collector side.

Assemble the circuit on a general-

purpose PCB and enclose in a small box. Keep the preset knob in the middle. In order to make it easy to plug the TUT, you can increase the number of test points as shown in the author's prototype in Fig. 2. ●

DIGITAL SOIL MOISTURE TESTER

■ T.K. HAREENDRAN

Here is a simple and compact digital soil moisture tester to check whether the soil is dry or wet. It can also be used to check the dryness or wetness of cotton, woolen and woven fabrics.

For indication, the tester (see Fig.1)

uses a number of LEDs driven by ubiquitous display driver IC LM3915 (IC1). The display panel shows the relative magnitude of conductance (reciprocal of resistance) between the two test probes when these are inserted in the soil.

The tester measures the dryness or wetness through soil resistance readings (0 to about 5 kilo-ohms) and

indicates in dot mode by sequential lighting of LED1 through LED9. The first LED (LED1) lights up when the conductance is high (resistance is almost nil).

Normally, the soil resistivity varies from 0 to about 5 kilo-ohms. So for calibration, connect a 5-kilo-ohm potentiometer between the two probes. Using trimpot VR1, set resistance to zero (minimum). LED1 should glow. Similarly, set resistance to 5 kilo-ohms (maximum). Now, LED9 should glow.

Assemble the circuit on a general-purpose PCB. After construction, enclose the tester (including battery) in a small plastic cabinet as shown in Fig. 2. The probes can be constructed from a pair of new injection needles. The needles should be firmly fixed about 2.5 cm apart on a piece of laminated plastic sheet. Use short length of a flat, twin wire-lead flexible cable for interconnection. After wiring the circuit, power the circuit using switch S1 and adjust zero-set trimpot (VR1) slowly such that LED1 lights up when probes are shorted. The tester is best powered by a compact 9V alkaline battery.

EFY note. For proper calibration, use a good soil moisture meter. ●

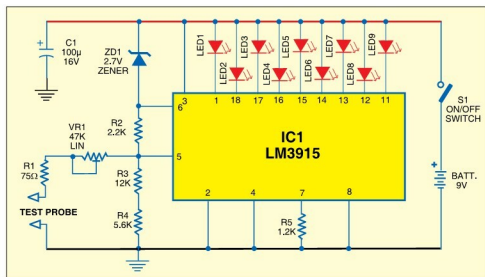


Fig. 1: Digital soil moisture tester

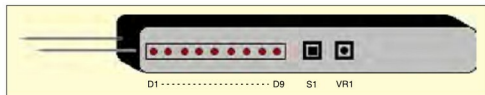


Fig. 2: Proposed cabinet with probes

WATER PUMP CONTROLLER

■ K.P. VISWANATHAN

Here is a simple circuit for controlling water level in an overhead tank. The main

components of this pump controller are a step-down transformer, a 24V AC double-changeover relay, two floats and two micro switches. Any available relay can be used irrespective

of its coil voltage. Of course, current rating of contacts should be taken into account according to the motor power.

